

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Easy

Question

A discount airline sells a certain number of tickets, x , for a flight for \$90 each. It sells the number of remaining tickets, y , for \$250 each. For a particular flight, the airline sold 120 tickets and collected a total of \$27,600 from the sale of those tickets. Which system of equations represents this relationship between x and y ?

Answer

- A. $\begin{cases} x + y = 120 \\ 90x + 250y = 27,600 \end{cases}$
- B. $\begin{cases} x + y = 120 \\ 90x + 250y = 120(27,600) \end{cases}$
- C. $\begin{cases} x + y = 27,600 \\ 90x + 250y = 120(27,600) \end{cases}$
- D. $\begin{cases} 90x = 250y \\ 120x + 120y = 27,600 \end{cases}$

Correct Answer: A

Rationale

Choice A is correct. The airline sold two types of tickets for this flight: x tickets at \$90 each and the remaining tickets, y , at \$250 each. Because the airline sold a total of 120 tickets for this flight, it must be true that $x + y = 120$. The amount, in dollars, collected from the sale of x tickets at \$90 each is represented by $90x$. The amount, in dollars, collected from the sale of the remaining y tickets at \$250 each is represented by $250y$. It is given that a total of \$27,600 was collected from the sale of all tickets. Therefore, it must also be true that $90x + 250y = 27,600$.

Choice B is incorrect. The total number of tickets sold is represented correctly as $x + y = 120$. The total amount, in dollars, collected from the sale of the x tickets at \$90 each and the remaining tickets, y , at \$250 has been correctly represented as $90x + 250y$. However, according to the information given, this total should be equal to 27,600, not $120(27,600)$ dollars. Choice C is incorrect. The total number of tickets sold has been correctly represented as $x + y$. However, according to the information given, this total should be equal to 120, not 27,600, as shown in choice C. The total amount, in dollars, collected from the sale of the x tickets at \$90 each and the remaining tickets, y , at \$250 has been correctly represented as $90x + 250y$. However, according to the information given, this total should be equal to 27,600, not $120(27,600)$ dollars. Choice D is incorrect. The two equations given in choice D have no meaning in this context.